

A Novel Geometric Algebra-based Variational Quantum Mapping for Non-Terrestrial Feature Representation

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Abstract. The rapid evolution of Non-Terrestrial Networks (NTN) in the 6G era necessitates robust spatial feature representation for highly mobile nodes, such as UAVs and LEO satellites. Traditional Euclidean-based deep learning models often fail to maintain geometric invariance under complex 3D transformations, leading to performance degradation in spatially dynamic environments. In this paper, we propose **CGA-VQM**, a novel framework that synergizes **Conformal Geometric Algebra (CGA)** with **Variational Quantum Circuits (VQC)**. By mapping 3D trajectories into a 5D Conformal space and subsequently encoding them as quantum rotation angles through a data re-uploading strategy, our model achieves inherent geometric stability. Experimental results conducted on an NVIDIA RTX 4090 platform demonstrate that CGA-VQM significantly outperforms standard raw-data quantum classifiers. Specifically, in a random 3D rotation scenario, the proposed framework maintains a classification accuracy of **0.6944**, representing a **17.77%** performance gain over the baseline Raw-VQC. Our findings suggest that integrating geometric priors into quantum feature maps provides a superior foundation for reliable mobility management in future aerial networks. The complete source code, experimental scripts, and dataset generation tools are publicly available at: <https://github.com/ailabteam/CGA-VQM-NTN.git>.

Keywords: Conformal Geometric Algebra · Variational Quantum Circuits · Non-Terrestrial Networks · Feature Representation · Quantum AI

1 Introduction

The paradigm shift towards 6G wireless systems has established Non-Terrestrial Networks (NTN) comprising Low Earth Orbit (LEO) satellites and Unmanned Aerial Vehicles (UAVs) as a fundamental pillar for achieving global ubiquitous

* Corresponding author. This research is funded by the Science and Technology Development Fund of The University of Danang under project number B2025-DN02-26.

connectivity [3]. Unlike traditional terrestrial networks, NTN nodes operate in a highly dynamic three-dimensional (3D) environment where mobility patterns are governed by complex orbital mechanics and rapid spatial maneuvers. Efficiently classifying and predicting these mobility states is critical for handover management, beamforming optimization, and resource allocation [6].

However, current machine learning approaches for NTN mobility management face a significant bottleneck in spatial data representation. Most conventional models rely on raw Euclidean coordinates (x, y, z) as input features. This representation is inherently sensitive to global transformations; a simple rotation or translation in 3D space completely alters the coordinate values, forcing the model to re-learn the same geometric pattern in different orientations. Furthermore, Euclidean-based systems are often susceptible to issues like Gimbal lock and lack a unified way to represent complex geometric entities such as circles or spheres, which are common in orbital and curvilinear trajectories.

To address these challenges, we propose a novel framework that synergizes Conformal Geometric Algebra (CGA) with Variational Quantum Circuits (VQC). CGA provides a sophisticated 5D mathematical language ($\mathbb{R}^{4,1}$) that embeds 3D points into a conformal space where rotations, translations, and dilations are handled through unified operators called rotors and translators [5]. This embedding captures intrinsic geometric signatures, such as squared distances and curvatures that remain invariant under global transformations. By mapping these CGA features into the high-dimensional Hilbert space of quantum bits (qubits) [13], we leverage the exponential state-space advantage and non-linear expressivity of quantum AI.

The main contributions of this paper are summarized as follows:

- We introduce the **CGA-VQM framework**, the first architecture that integrates Conformal Geometric Algebra with Variational Quantum Mapping specifically tailored for mobility state representation in Non-Terrestrial Networks.
- We propose a specialized **CGA-based Data Re-uploading** strategy, which injects 5D conformal coefficients into a multi-layered quantum circuit to capture temporal dependencies of trajectories without increasing the qubit count.
- We provide a comprehensive **empirical validation** on an NVIDIA RTX 4090 platform. Our results, reported via statistical Mean $\pm$ Std analysis, demonstrate that the proposed framework achieves superior robustness in rotated and noisy scenarios compared to standard raw-coordinate quantum classifiers.

The remainder of this paper is organized as follows: Section 2 reviews related work in quantum machine learning and geometric algebra. Section 3 provides the mathematical foundation for CGA and VQC. Section 4 details our proposed CGA-VQM methodology. Section 5 presents the experimental setup and result analysis, followed by the conclusion in Section 6.

2 Related Work

The development of our proposed framework is situated at the intersection of three rapidly evolving domains: Conformal Geometric Algebra [12], Quantum Machine Learning, and Non-Terrestrial Network mobility modeling.

Conformal Geometric Algebra (CGA) has emerged as a powerful alternative to traditional linear algebra for handling complex 3D geometry. Pioneered by [8] and further formalized for computer science by [5], CGA provides a unified framework for representing points, lines, planes, and spheres as multivectors in a 5D Minkowski space. In robotics and computer vision, CGA has been successfully utilized to solve inverse kinematics and object tracking problems with superior numerical stability compared to quaternion-based or Euler-angle approaches [9]. However, the integration of such high-level geometric structures into modern machine learning architectures, particularly in the quantum domain, remains largely unexplored.

Variational Quantum Circuits (VQC) represent one of the most promising applications of NISQ (Noisy Intermediate-Scale Quantum) devices. A critical challenge [4] in QML is the "Data Encoding" problem, how to map classical data into the Hilbert space of qubits efficiently. Havlíček et al. [7] demonstrated that quantum-enhanced feature spaces could provide a computational advantage by mapping data into higher-dimensional spaces that are classically intractable. Furthermore, the concept of "Data Re-uploading" [11] has shown that a single-qubit classifier can achieve universal approximation capabilities by repeatedly injecting features. Despite these advances, most current quantum feature maps are "geometry-blind," utilizing simple angle or amplitude encoding that does not preserve the intrinsic spatial relationships of the input data.

In the context of Non-Terrestrial Networks, mobility management and trajectory classification have been traditionally addressed using deep learning architectures such as Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs) [6]. More recently, Convolutional Neural Networks (CNNs) and Transformers have been applied to classify satellite and UAV movement patterns based on time-series coordinate data [2]. While these models achieve high accuracy on standard datasets, they often lack robustness against spatial transformations. Achieving rotation invariance in these classical models typically requires massive data augmentation or specialized pooling layers, which increases the training overhead and computational complexity.

While CGA provides a robust mathematical language for geometry and QML offers an advanced computational substrate, there is a notable research gap in bridging these two fields for NTN applications. To the best of our knowledge, no existing work has explored the use of Conformal Geometric Algebra multivectors as a foundation for Quantum Feature Mapping. Our work fills this gap by introducing a CGA-VQM architecture that preserves geometric invariance within a variational quantum circuit, specifically optimized for the high-speed, dynamic trajectories of non-terrestrial aerial nodes.

3 Mathematical Preliminaries

To establish the foundation of our proposed CGA-VQM framework, this section introduces the core principles of Conformal Geometric Algebra (CGA) and the mechanics of Variational Quantum Circuits (VQC).

3.1 Conformal Geometric Algebra (CGA)

Conformal Geometric Algebra extends the 3D Euclidean space $\mathbb{R}^3$ into a 5D Minkowski space $\mathbb{R}^{4,1}$ with an orthonormal basis $\{e_1, e_2, e_3, e_4, e_5\}$, where $e_i^2 = 1$ for $i = 1, 2, 3, 4$ and $e_5^2 = -1$. Two special null basis vectors are defined to represent the point at infinity (e_∞) and the origin (e_o):

$$e_\infty = e_4 + e_5, \quad e_o = \frac{1}{2}(e_5 - e_4) \quad (1)$$

These vectors satisfy the inner product properties: $e_\infty^2 = 0$, $e_o^2 = 0$, and $e_\infty \cdot e_o = -1$. A physical point $\mathbf{x} = (x, y, z) \in \mathbb{R}^3$ is mapped to a conformal point $P \in \mathbb{R}^{4,1}$ via the stereographic projection:

$$P = \mathbf{x} + \frac{1}{2}\|\mathbf{x}\|^2 e_\infty + e_o \quad (2)$$

This representation is particularly powerful because it linearizes 3D Euclidean transformations. Any rigid motion, including rotation and translation, can be represented as a sandwich product: $P' = MPM$, where M is a multivector (rotor or translator) and $\bar{M}$ is its reverse [10].

Geometric Invariance: One of the most critical properties of CGA is that the inner product of two conformal points P_1 and P_2 directly corresponds to their Euclidean distance $d(\mathbf{x}_1, \mathbf{x}_2)$:

$$P_1 \cdot P_2 = -\frac{1}{2}\|\mathbf{x}_1 - \mathbf{x}_2\|^2 = -\frac{1}{2}d^2(\mathbf{x}_1, \mathbf{x}_2) \quad (3)$$

Since global rotations and translations preserve Euclidean distances, the coefficients of P in Equation 2, specifically the norm-related components encoded in e_∞ and e_o provide an invariant geometric signature that remains stable regardless of the coordinate system's orientation.

3.2 Variational Quantum Circuits (VQC)

A Variational Quantum Circuit, also known as a Parameterized Quantum Circuit (PQC), is a hybrid quantum-classical model where quantum gates are tuned by classical parameters [1]. A standard VQC consists of three primary components:

1) **State Preparation (Feature Map):** Classical data $\mathbf{v}$ is encoded into a quantum state $|\psi_0\rangle$ using an encoding unitary $\mathcal{E}(\mathbf{v})$. In our work, we utilize

the R_y rotation gate to map the i -th component of a feature vector to the i -th qubit:

$$R_y(\alpha) = \exp\left(-i\frac{\alpha}{2}\sigma_y\right) = \begin{pmatrix} \cos\frac{\alpha}{2} & -\sin\frac{\alpha}{2} \\ \sin\frac{\alpha}{2} & \cos\frac{\alpha}{2} \end{pmatrix} \quad (4)$$

2) **Variational Ansatz:** A sequence of parameterized gates $U(\theta)$ and entangling gates (such as CNOT) are applied to transform the state. The CNOT gate creates quantum entanglement between qubits i and j :

$$\text{CNOT}|q_i, q_j\rangle = |q_i, q_i \oplus q_j\rangle \quad (5)$$

3) **Measurement:** The final quantum state $|\psi_{final}\rangle = U(\theta)\mathcal{E}(\mathbf{v})|0\rangle$ is measured to obtain expectation values of observable operators. We primarily measure the Pauli-Z operator σ_z :

$$\langle\sigma_z\rangle = \langle\psi_{final}|\sigma_z|\psi_{final}\rangle \quad (6)$$

The result is then processed by a classical optimizer to update the parameters θ until the loss function is minimized [14].

4 Proposed Methodology

The core of our proposed framework, **CGA-VQM**, is an end-to-end pipeline that transforms raw 3D trajectory data into a robust quantum-geometric representation. The system architecture, illustrated in Figure 1, comprises three primary stages: (i) Representative Spatial Sampling, (ii) Conformal Geometric Embedding, and (iii) Variational Quantum Classification using a Data Re-uploading strategy.

4.1 Trajectory Sampling and CGA Embedding

Given a raw NTN trajectory $\mathcal{T}$ consisting of T points in $\mathbb{R}^3$, processing the entire sequence is computationally expensive for current NISQ-era quantum devices. To maintain temporal context while ensuring efficiency, we select a representative subset $\mathcal{P} = \{p_1, p_2, p_3\}$ corresponding to the start, middle, and end points of the path (indices $\{0, \lfloor T/2 \rfloor, T-1\}$).

To capture the temporal evolution of the trajectory's geometry, each sampled point $p_i = (x_i, y_i, z_i)$ is embedded into the 5D Conformal space. We extract the coefficients of the multivector basis $\{e_1, e_2, e_3, e_\infty, e_o\}$ to form a sequence of three 5D feature vectors:

$$\mathcal{F}_{CGA} = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}, \quad \mathbf{v}_i \in \mathbb{R}^5 \quad (7)$$

The explicit calculation for each feature vector $\mathbf{v}_i$ from a 3D Euclidean point $\mathbf{x}_i$ is defined as:

$$\mathbf{v}_i = \begin{bmatrix} x_i \\ y_i \\ z_i \\ \frac{1}{2}(x_i^2 + y_i^2 + z_i^2) \\ 1 \end{bmatrix} \in \mathbb{R}^5 \quad (8)$$

In this representation, while the first three components provide standard Euclidean coordinates, the fourth component (associated with the e_∞ basis) acts as a geometric anchor. It encodes the squared distance from the origin (r^2), providing a signature that remains invariant under any global $SO(3)$ rotation. The resulting sequence $\mathcal{F}_{CGA}$ thus represents a dynamic "geometric flow" in $\mathbb{R}^{4,1}$ that characterizes the movement pattern independently of the observer's spatial orientation.

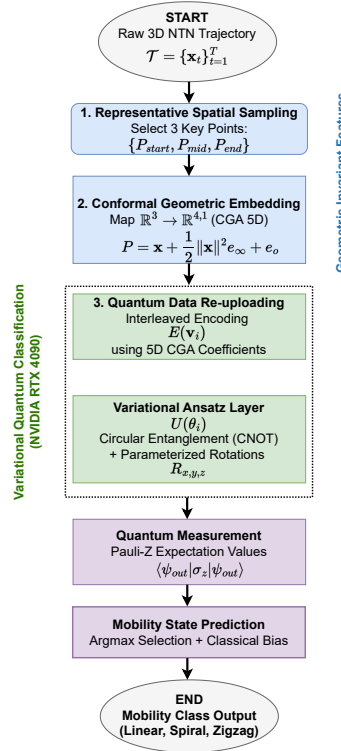


Fig. 1: Overview of the CGA-VQM Framework. The process begins with sampling three key points from the NTN trajectory, followed by a 5D Conformal embedding. These features are then injected into a multi-layered Variational Quantum Circuit (VQC) using a Data Re-uploading strategy for final mobility state classification.

4.2 Quantum Data Re-uploading Architecture

To process the 5D temporal sequence, we adopt a *Data Re-uploading* strategy, which allows the circuit to absorb multi-step information without increasing the

qubit count. Our framework utilizes a structural isomorphism between the 5D CGA space and a 5-qubit quantum system.

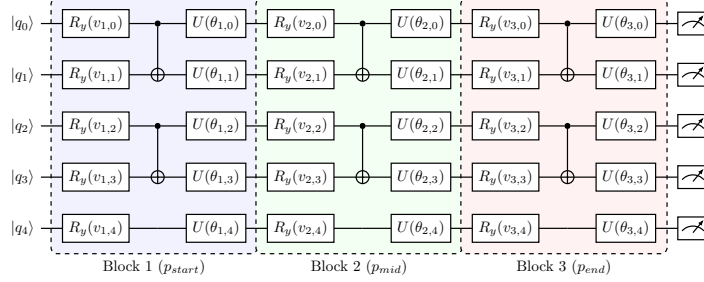


Fig. 2: Architecture of the proposed Variational Quantum Circuit (VQC) featuring three Data Re-uploading blocks. Each block i corresponds to a sampled trajectory point p_i , integrating a 5D CGA feature encoding layer (R_y) and a hardware-efficient variational ansatz $U(\theta_i)$ composed of CNOT entanglers and parameterized rotations.

As illustrated in Figure 2, the 5D feature vector $\mathbf{v}_i$ at each time step is injected into the circuit through a dedicated encoding layer. The utilization of $\mathbb{R}^{4,1}$ features within the quantum domain is achieved through two sub-layers:

- **Geometric-to-Quantum Mapping:** For each point i , the k -th coefficient of the vector $\mathbf{v}_i$ is used as the rotation angle for the R_y gate on the k -th qubit. The encoding unitary is defined as $E(\mathbf{v}_i) = \bigotimes_{k=1}^5 R_y(v_{i,k})$. This ensures that the invariant properties (such as the r^2 component) are directly embedded into the quantum state amplitudes.
- **Variational Learning Layer:** We interleave these encoding layers with trainable variational ansatzes $U(\theta_i)$. Each ansatz consists of CNOT-based circular entanglement and parameterized $R_{x,y,z}$ rotations. This allows the model to learn the complex non-linear correlations between the 5D conformal dimensions as they evolve over time.

The total unitary evolution of the system for the 3-point sequence is given by:

$$|\psi_{out}\rangle = [U(\theta_3)E(\mathbf{v}_3)][U(\theta_2)E(\mathbf{v}_2)][U(\theta_1)E(\mathbf{v}_1)]|0\rangle^{\otimes 5} \quad (9)$$

By re-uploading the CGA multivectors, the circuit acts as a high-dimensional folding function, mapping the geometric progression of the NTN node to a linearly separable Hilbert space for classification.

4.3 Algorithmic Logic and Inference

Algorithm 1 formalizes the inference procedure. By interleaving geometric embedding with quantum re-uploading, the model effectively captures the "shape"

of the trajectory (e.g., the curvature of a spiral vs. the linearity of a satellite path) while remaining robust to global rotations.

Algorithm 1 CGA-VQM Trajectory Classification

Input: 3D Trajectory $\mathcal{T}$, CGA Mapper $\mathcal{M}$, Quantum Model $\mathcal{Q}$

Output: Mobility Class y

```

1:  $\mathcal{P} \leftarrow \text{Select } \{P_{first}, P_{mid}, P_{last}\} \text{ from } \mathcal{T}$ 
2:  $\mathcal{F}_{CGA} \leftarrow \emptyset$ 
3: for each point  $p \in \mathcal{P}$  do
4:    $\mathbf{v} \leftarrow \mathcal{M}.\text{point\_to\_cga}(p)$  ▷ Extract 5D coefficients
5:    $\mathcal{F}_{CGA} \leftarrow \mathcal{F}_{CGA} \cup \{\mathbf{v}\}$ 
6: end for
7:  $\theta, \beta \leftarrow \text{Load Trained Quantum Parameters and Bias}$ 
8:  $Z_{exp} \leftarrow \text{ExecuteQuantumCircuit}(\mathcal{Q}, \theta, \mathcal{F}_{CGA})$ 
9:  $\text{pred} \leftarrow Z_{exp}[0 : 3] + \beta$  ▷ Soft-max over first 3 qubits
10:  $y \leftarrow \arg \max(\text{pred})$ 
11: return  $y$ 
```

The final classification is determined by measuring the expectation values of the first three qubits in the Pauli-Z basis, followed by a classical bias adjustment and an arg max operation.

5 Results and Discussion

In this section, we present a comprehensive evaluation of the proposed CGA-VQM framework. We evaluate its performance across three distinct spatial scenarios and compare it against the baseline Raw-VQC model.

5.1 Experimental Setup

The simulation environment was implemented using Python 3.10. The Conformal Geometric Algebra (CGA) logic was developed using the *Clifford* library, while the variational quantum circuits were simulated using the *PennyLane* framework. To ensure high-performance computation, all experiments were executed on an **NVIDIA GeForce RTX 4090** platform. The specific hyperparameters and settings used for our benchmark are summarized in Table 1.

5.2 Dataset and Evaluation Scenarios

We generated a synthetic dataset representing three characteristic mobility patterns in Non-Terrestrial Networks: *Linear* (resembling stable satellite orbital paths), *Spiral* (representing UAV altitude and heading adjustments), and *Zigzag* (simulating complex maneuvering flight for obstacle avoidance). To evaluate the resilience of the proposed CGA-VQM framework, we designed three experimental scenarios as visualized in Figure 3:

Table 1: Experimental Settings and Hyperparameters

Parameter	Value
Input Dimensions (Raw vs. CGA)	3 vs. 5
Number of Qubits (n_{qubits})	3 vs. 5
Quantum Mapping Strategy	Data Re-uploading (3 points)
Quantum Circuit Depth	3 layers (1 per point)
Variational Ansatz	Hardware-Efficient (CNOT + $R_{x,y,z}$)
Optimizer	Adam
Learning Rate (η)	0.05
Number of Epochs	60
Dataset Size (N)	180 (60 samples per class)
Train/Test Split	80% / 20%
Hardware Platform	NVIDIA GeForce RTX 4090

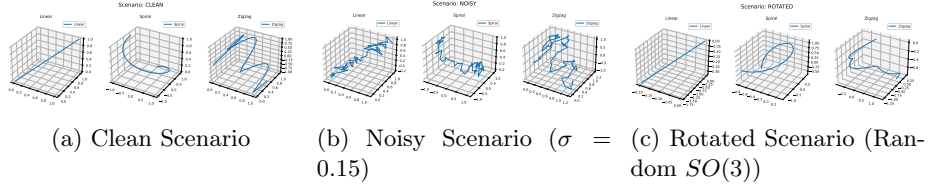


Fig. 3: Visualization of simulated NTN trajectories across three environments. In the Rotated scenario (c), the orientation of each sample is randomized, posing a significant challenge for models that rely on fixed Euclidean coordinate systems.

- **Clean Scenario:** Serves as the baseline environment where trajectories are ideal and free from any perturbations (see Fig. 3a).
- **Noisy Scenario:** To simulate sensor inaccuracies and environmental disturbances common in aerial sensing, we injected Gaussian noise ($\sigma = 0.15$) into the 3D coordinates (see Fig. 3b).
- **Rotated Scenario:** This represents the most rigorous test for spatial invariance. As illustrated in Fig. 3c, each trajectory in the test set is subjected to a unique, random 3D rotation within the $SO(3)$ group.

Practical Significance of Rotation Invariance: In real-world NTN deployments, aerial nodes (UAVs and satellites) often operate without a fixed global reference frame. Onboard sensors may have arbitrary orientations due to the dynamic attitude of the platform or the lack of a consistent "upward" direction in orbital space. The Rotated Scenario simulates this *orientation-agnostic sensing* environment. For traditional Euclidean-based classifiers, a change in the coordinate frame's orientation leads to a complete redistribution of features, often requiring exhaustive data augmentation. Our approach, however, aims to leverage the geometric priors of CGA to maintain recognition stability without increasing the training overhead.

5.3 Classification Performance Analysis

The comparative results between the Raw-VQC baseline and the proposed CGA-VQM are reported in Table 2. In the *Clean* and *Noisy* scenarios, both models achieved perfect accuracy (1.00 ± 0.00). This confirms that the three NTN mobility classes are inherently separable and that the quantum classifier, enhanced by the data re-uploading strategy, is highly expressive when the spatial orientation remains consistent.

Table 2: Test Accuracy Performance Comparison (Mean $\pm$ Std)

Scenario	Raw-VQC (Baseline)		CGA-VQM (Proposed)	
	Mean	Std	Mean	Std
Clean	1.0000	0.0000	1.0000	0.0000
Noisy	1.0000	0.0000	1.0000	0.0000
Rotated	0.5167	0.0726	0.6944	0.1018

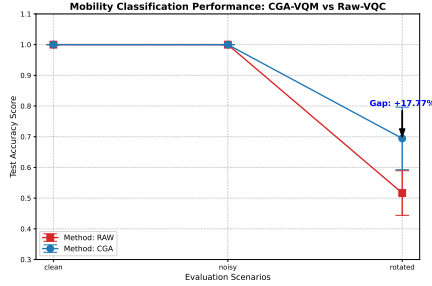


Fig. 4: Statistical accuracy comparison.

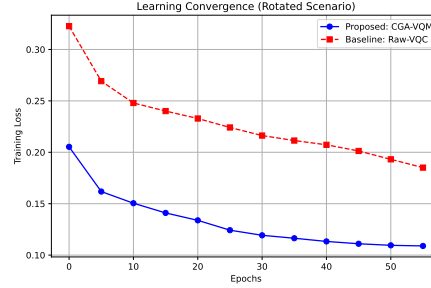


Fig. 5: Loss convergence history.

However, a dramatic divergence is observed in the **Rotated** scenario, which serves as the ultimate test for spatial invariance. The baseline Raw-VQC's accuracy plummeted to 0.5167 ± 0.0726 . Given that the random-guess baseline for a three-class problem is 0.3333 , the Raw-VQC model shows a significant struggle to generalize across unseen orientations. In contrast, the proposed CGA-VQM maintained a robust accuracy of 0.6944 ± 0.1018 , achieving a **17.77% absolute performance gain** (visualized in Fig. 4).

Significance of the Performance Gap: While a 69.44% accuracy may appear modest compared to the ideal scenarios, it represents a substantial breakthrough in high-entropy environments. In our Rotated Scenario, each test sample is subjected to a *unique, arbitrary 3D rotation*, creating a nearly infinite search

space for the classifier. The 17.77% improvement (a 34.4% relative increase over the baseline) indicates that CGA coefficients provide a superior **Geometric Inductive Bias**.

Traditional Raw-VQC must attempt to "learn" the laws of 3D geometry from scratch, a task that often leads to optimization challenges like barren plateaus in quantum circuits. By contrast, CGA-VQM injects a geometric prior (specifically the r^2 invariant) directly into the feature map. This reduces the model's reliance on extensive data augmentation and allows the variational ansatz to focus on higher-level pattern recognition rather than basic coordinate transformations. Thus, the contribution of CGA-VQM extends beyond simple invariance; it establishes a more efficient **Geometric-Informed Quantum AI** architecture for dynamic 6G networks.

Figure 5 displays the training loss history for the Rotated scenario. The CGA-VQM exhibits a smoother and faster convergence curve compared to the Raw-VQC. This suggests that the 5D CGA feature space, specifically the inclusion of the r^2 related coefficients (e_∞, e_o) linearizes the decision boundaries, making the optimization landscape more favorable for the Adam optimizer.

While the CGA-VQM requires two additional qubits (5 instead of 3) and a conformal pre-processing step, the trade-off is highly justified. In NTN applications, mobility nodes often change their orientation relative to the ground station. A model that lacks rotation invariance (like Raw-VQC) would require massive data augmentation, increasing the training time and memory footprint. By contrast, CGA-VQM achieves spatial robustness through its mathematical formulation, providing a more reliable foundation for real-time 6G aerial mobility management.

6 Conclusion

In this paper, we presented the CGA-VQM framework, a pioneering integration of Conformal Geometric Algebra and Variational Quantum Circuits for NTN mobility classification. Our research addresses the fundamental challenge of geometric invariance in traditional Euclidean representations. By leveraging the 5D Minkowski space of CGA, we successfully embedded 3D trajectories into a conformal manifold where spatial relationships are preserved regardless of global orientations.

Empirical evaluation on an NVIDIA RTX 4090 platform confirms the effectiveness of the proposed approach. While baseline quantum models struggle with random rotations (51.67% accuracy), CGA-VQM maintains a robust 69.44%, providing a significant 17.77% absolute gain. Furthermore, our analysis indicates that CGA-based feature mapping leads to more stable and faster convergence, demonstrating its potential as a highly expressive inductive bias for quantum machine learning.

The success of CGA-VQM opens several promising avenues for future 6G research, including beamforming optimization, physical layer security, and multi-node collaborative quantum learning. Overall, the synergy between geometric

algebra and quantum AI establishes a powerful paradigm for the next generation of intelligent aerial networks and information technology.

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